

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 7 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

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|--|--|
| 1. Singular value C | (A) A column times a row, $\mathbf{uv}^T$ — a whole matrix built from two vectors. |
| 2. Singular vectors E | (B) The factorization $A = U\Sigma V^T$ (rotate–stretch–rotate) that <i>every</i> matrix has. |
| 3. Singular Value Decomposition B | (C) A number $\sigma \geq 0$: how far A stretches along a singular direction (a half-width of the ellipse). |
| 4. Outer product A | (D) A square matrix with perpendicular unit columns, so $Q^T Q = I$ and $Q^{-1} = Q^T$. |
| 5. Rank-one matrix F | (E) The perpendicular unit inputs $\mathbf{v}_i$ and outputs $\mathbf{u}_i$ with $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$. |
| 6. Orthogonal matrix D | (F) A matrix whose every column is a multiple of one vector (like $\mathbf{uv}^T$). |
| 7. Principal component H | (G) The total $\sigma_1^2 + \sigma_2^2 + \cdots$, equal to the sum of the squares of all entries of A . |
| 8. Energy of a matrix G | (H) A top singular direction of centered data — the direction of greatest spread. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- The singular values σ of a matrix are **(A) always ≥ 0 ←** (B) sometimes negative (C) the eigenvalues of A (D) always 1
- To find the singular values of A , use the eigenvalues of (A) A (B) $A + A^T$ **(C) $A^T A$ ←** (D) A^{-1}
- A singular value σ_i and the eigenvalue λ_i of $A^T A$ are related by (A) $\sigma = \lambda$ **(B) $\sigma = \sqrt{\lambda}$ ←** (C) $\sigma = \lambda^2$ (D) $\sigma = 1/\lambda$
- In $A = U\Sigma V^T$, all the stretching is carried by (A) U (B) V **(C) Σ ←** (D) V^T
- An orthogonal matrix Q satisfies (A) $Q^2 = I$ **(B) $Q^T Q = I$, so $Q^{-1} = Q^T$ ←** (C) $\det Q = 0$ (D) $Q = Q^T$ always
- To compress an image with the SVD, you keep the rank-one layers with the (A) smallest σ **(B) largest σ ←** (C) largest $\mathbf{u}$ (D) negative σ

Teacher Note

Item 1: singular values are always ≥ 0 (they are $\sqrt{\lambda}$, lengths of the ellipse's half-axes). Item 2: σ 's come from the eigenvalues of the symmetric $A^T A$. Item 3: $\sigma = \sqrt{\lambda}$. Item 4: U and V^T are rotations (they only turn); Σ is the only factor that stretches. Item 5: orthogonal means $Q^T Q = I$, so the inverse is free, $Q^{-1} = Q^T$. Item 6: compression keeps the biggest- σ layers (they hold the most energy).

Part C — Short Answer & Computation (35 pts)

Show your work.

- Singular values from geometry. (a) **A diagonal matrix stretches by its diagonal entries, so the singular values are $\sigma_1 = 4$, $\sigma_2 = 3$ (largest first). The unit circle becomes an ellipse with half-widths 4 and 3.** (b) **The columns of $S = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$ are $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ (length 1) and $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$ (length 2), and they are perpendicular, so $\sigma_1 = 2$, $\sigma_2 = 1$.**
- Singular values of $A = \begin{bmatrix} 2 & 2 \\ 1 & -2 \end{bmatrix}$ via $A^T A$. $A^T A = \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix}$ (entries are the column dot products). $\det(A^T A - \lambda I) = (5 - \lambda)(8 - \lambda) - 4 = \lambda^2 - 13\lambda + 36 = (\lambda - 9)(\lambda - 4) = 0$, so $\lambda = 9, 4$. Therefore $\sigma_1 = \sqrt{9} = 3$ and $\sigma_2 = \sqrt{4} = 2$.
- Output vector $\mathbf{u}_1 = \frac{A\mathbf{v}_1}{\sigma_1}$. $A\mathbf{v}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 6 \\ -3 \end{bmatrix}$. Divide by $\sigma_1 = 3$: $\mathbf{u}_1 = \frac{1}{3\sqrt{5}} \begin{bmatrix} 6 \\ -3 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. **Unit check:** $|\mathbf{u}_1|^2 = \frac{1}{5}(2^2 + (-1)^2) = \frac{5}{5} = 1$. ✓
- Outer product and rank one. (a) $\mathbf{u}\mathbf{v}^T = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix}$. (b) **Both columns are multiples of $\mathbf{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ (namely $2\mathbf{u}$ and $\mathbf{u}$), so every column points along one line — the matrix has rank one.**
- Rank-one approximation and energy of $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$. (a) $A_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T = 6 \cdot \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$. (b) **Total energy $= \sigma_1^2 + \sigma_2^2 = 36 + 4 = 40$ (equal to $4^2 + 2^2 + 2^2 + 4^2 = 40$). A_1 keeps $\frac{36}{40} = 90\%$ of the energy.**
- Orthogonal matrix $Q = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$. (a) **The columns are $\frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\frac{1}{5} \begin{bmatrix} -4 \\ 3 \end{bmatrix}$: each has length $\frac{1}{5} \sqrt{9+16} = 1$, and their dot product is $\frac{1}{25}(-12+12) = 0$. So $Q^T Q = I$ (1's on the diagonal, 0's off).** (b) $Q^{-1} = Q^T = \frac{1}{5} \begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix}$ (the inverse is free). (c) $Q\mathbf{x} = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, which has length $\sqrt{9+16} = 5$ — the same length as $\mathbf{x}$, because an orthogonal matrix is a rigid motion (rotation/reflection) that preserves length.
- PCA of $(5, 7), (7, 5), (3, 1), (1, 3)$. (a) **Mean $= (4, 4)$; centered points $(1, 3), (3, 1), (-1, -3), (-3, -1)$.** (b) $A^T A = \begin{bmatrix} 20 & 12 \\ 12 & 20 \end{bmatrix}$ (column dot products of the centered data). (c) $\lambda = 20 \pm 12 = 32, 8$; the top eigenvector is $\mathbf{PC}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. (d) $\mathbf{PC}_1$ captures $\frac{32}{32+8} = \frac{32}{40} = 80\%$ of the spread.

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

- Why every matrix has singular values. (a) $A^T A$ is symmetric because $(A^T A)^T = A^T (A^T)^T = A^T A$ — it equals its own transpose no matter what shape A is. By the spectral theorem a symmetric matrix has real eigenvalues and perpendicular eigenvectors; and each eigenvalue satisfies $\lambda = \frac{|A\mathbf{v}|^2}{|\mathbf{v}|^2} \geq 0$ (a length squared over a length squared), so $\lambda \geq 0$. (b) Since every $\lambda \geq 0$, its square root $\sigma = \sqrt{\lambda}$ is a real number ≥ 0 . These σ 's are the singular values — and because $A^T A$ exists and is symmetric for any matrix, every matrix has them (unlike eigenvectors, which can be complex or too few).

2. Rotate–stretch–rotate and orthogonal matrices. (a) **Reading** $A = U\Sigma V^T$ **right to left:** V^T *rotates* the input frame onto the axes, Σ *stretches* along those axes by the singular values $\sigma_1, \sigma_2, \dots$, and U *rotates* the result to the output frame. U and V^T are orthogonal, so they only turn (no resizing); all the stretching is packed into the diagonal Σ . (b) $Q^T Q = I$ says Q^T times Q gives the identity, which is exactly what it means for Q^T to be the inverse: $Q^{-1} = Q^T$, free of any computation. Length is preserved because $|Q\mathbf{x}|^2 = (Q\mathbf{x})^T(Q\mathbf{x}) = \mathbf{x}^T Q^T Q \mathbf{x} = \mathbf{x}^T \mathbf{x} = |\mathbf{x}|^2$, so $|Q\mathbf{x}| = |\mathbf{x}|$.

Teacher Note

Scoring Part D (5 pts each). D1: 3 pts (a) $A^T A$ symmetric for any $A \Rightarrow$ real, perpendicular eigenvectors and $\lambda \geq 0$ (via $\lambda = |A\mathbf{v}|^2/|\mathbf{v}|^2$); 2 pts (b) $\sigma = \sqrt{\lambda} \geq 0$ real, so every matrix has singular values. D2: 3 pts (a) V^T/U rotate, Σ stretches, all stretching in Σ ; 2 pts (b) $Q^T Q = I \Rightarrow Q^{-1} = Q^T$, and $|Q\mathbf{x}|^2 = \mathbf{x}^T Q^T Q \mathbf{x} = |\mathbf{x}|^2$. Accept equivalent wording throughout.